LESSON PLAN

AFCS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Infra-red homing, its block diagram and advantages and limitations of IR guidance Demonstration on Helmet mounted target designation system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Infra-red homing, its block diagram and advantages and limitations of IR guidance Demonstration on Helmet mounted target designation system Film on Infra-red homing techniques
5. Specific Objectives	22.1.1.1Trainee will be able to understand the basic of guided weapon systems and types of targets.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration on Helmet mounted target designation system (HMTDS) including safety precautions regarding laser safety and instructor supervision of trainee interaction with the system interface.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the two primary types of guided weapon systems?
(b) Define the term 'Infra-red homing' in one sentence.
(c) List three common sources of heat that can act as targets for IR seekers.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of Infra-red homing • Block diagram of IR guidance system • Advantages of IR guidance • Limitations of IR guidance	L & Ds	PPT, IB	15 min
(b)	• Helmet mounted target designation system overview • Operational sequence of HMTDS • Integration of IR seeker with HMTDS • Safety protocols for laser designation	L & Ds	PPT, PDF	20 min
(c)	• Comparison of IR vs Radar homing • Operational scenarios for IR weapons • Maintenance considerations for IR seekers • Summary of IR guidance constraints	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the block diagram of an IR guidance system differs from a radar homing system.
(b) Describe one major limitation of IR guidance and how HMTDS mitigates it.
(c) Identify two safety precautions required during the demonstration of the Helmet mounted target designation system.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational limitations of Infra-red homing with the low altitude limitations of Radar homing to be discussed in Lesson Plan 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Radar homing and sources of error and low altitude limitations.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min